ROBOT I SENSORS

Rep
- /cmd_vel

Publica
- /scan
- /odom
- /tf
- /camera/image/compressed

**

Publica **/cmd_vel** segons la fase:
- free_space_explorer_node
- botticelli_interface_node
- drawing_navigator_node
- drawing_follower_node

SLAM / LOCALITZACIÓ EXTERNA

1) Mode 1 - SLAM

slam_gmapping

Rep:
- /scan
- /odom
- /tf

Publica:
- /map
- /tf map -> odom

(Construeix el mapa)

free_space_explorer_node

Rep:
- /scan
- /map
- /tf map -> base

Publica:
- /cmd_vel
- /debug_state
- /botticelli/drawing_center
- /botticelli/free_area_info
- /botticelli/navigation_done
- /botticelli/current_phase

Genera:
- mapa.yaml
- mapa.pgm
- mapa_features.txt

(Explora, evita obstacles,guarda el mapa i registra features)

2) Mode 2 - Mapa existent

map_server

Rep:
- fitxer yaml

Publica:
- /map

(Carrega mapa existent)

amcl

Rep:
- /map
- /scan
- /tf
- /odom
- /initialpose
- /global_localization

Publica:
- /amcl_pose
- /particlecloud
- /tf map -> odom

(Localitza el robot en un mapa existent)

INTERFICIE

botticelli_interface_node

Rep:
- /contour_info
- /drawing_path
- /drawing_ready
- /drawing_finished
- /drawing_progress
- /boticelli/current_phase

Publica:
- /cmd_vel
- /capture_contour
- /drawing_dimensions
- /go_to_drawing
- /start_drawing

Servei:
- /global_localization

(Coordina totes les fases)

VISIÓ, PLANNER, NAVIGATOR I FOLLOWER

vision_node

Rep:
- /camera/image/compressed
- /capture_contour

Publica:
- /contour_points
- /contour_info

(Detecta el contorn del dibuix)

drawing_planner_node

Rep:
- /contour_points
- /drawing_dimensions
- /tf map -> base_footprint

Publica:
- /drawing_path
- /drawing_marker

(Converteix contorn en trajectòria)

drawing_navigator_node

Rep:
- /drawing_path
- /go_to_drawing
- /tf map -> base_footprint

Publica:
- /cmd_vel
- /drawing_ready
- /navigator_goal_marker

(Porta el robot al punt inicial per dibuixar)

drawing_follower_node

Rep:
- /drawing_path
- /start_drawing
- /scan
- /tf map -> base_footprint

Publica:
- /cmd_vel
- /executed_path
- /lookahead_marker
- /drawing_progress
- /drawing_finished

(Executa la trajectòria del dibuix)

RViz

Rep:
- /map
- /scan
- /robot_description
- /tf
- /drawing_path
- /executed_path
- /drawing_marker
- /navigator_goal_marker
- /lookahead_marker

(Visualització)

LLEGENDA:

Bloc funcional

Nodes externs ROS

Nodes propis

Robot / visualització